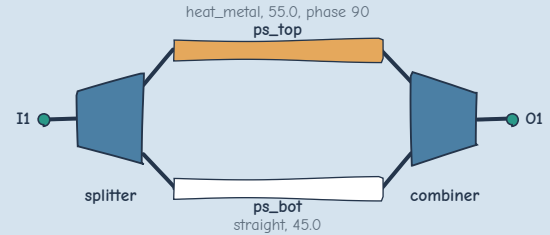
a the netlist (JSON)

```
{
  "netlist": {
    "instances": {
      "splitter": "mmi",
      "combiner": "mmi",
      "ps_top": {
        "component": "straight_heat_metal",
        "settings": {"length": 55.0, "phase_shift": 90}
      },
      "ps_bot": {
        "component": "straight",
        "settings": {"length": 45.0}
      }
    },
    "connections": {
      "splitter,O1": "ps_top,I1",
      "splitter,O2": "ps_bot,I1",
      "ps_top,O1": "combiner,O1",
      "ps_bot,O1": "combiner,O2"
    },
    "ports": {
      "I1": "splitter,I1",
      "O1": "combiner,I1"
    }
  },
  "models": {...}
}
```

b the circuit (graph)



c the chip layout (not scored)

